

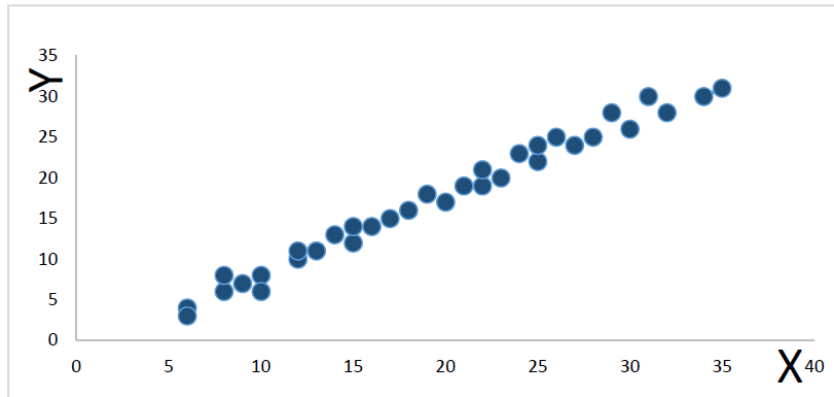
STA201: Elements of Statistics and Probability

Lecture 8

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2 July, 2024

Regression Analysis

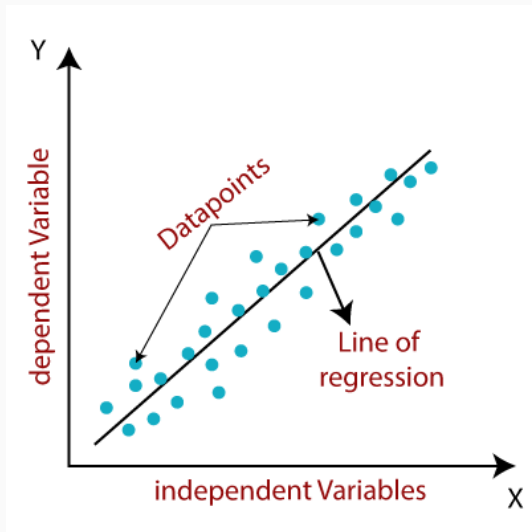
Motivation



Motivation Cont...

- The above scatter plot shows strong correlation between X and Y.
- If we want to develop a **mathematical model or equation** between X and Y, we need to fit a line through these sample points.
- The line should **best describes the relationships** between the variables.
- This line is known as **regression line**.

Motivation Cont...



Correlation Vs Regression

Correlation

- We consider two variables symmetrically. This means we can interchange between variables.
- It can't tell us the amount of change in one variable to a unit change in other variable and hence can't predict future values.
- There may be nonsense relation in correlation.
- Correlation is not very useful for further mathematical treatment.
- It indicates only linear relationship between variables.

Regression

- There is an asymmetry in the way dependent and independent variables are treated. Here the dependent variable is random and independent variable is fixed and can't be interchanged.
- It tells us the amount of change in the dependent variable to a unit change in the independent variable, hence can predict future values.
- There is no such relation in regression.
- Regression is widely used for further mathematical treatment.
- It indicates any type of relationship.

Regression Analysis

- Regression analysis is a mathematical measure of **the relationship between one dependent variable with one/more independent variables.**
- Regression is used to denote estimation or prediction of the average value of dependent variable for a specified value of the independent variable.
- In regression analysis, there are **two types** of variables.
 - **Dependent Variable (Y)**
 - **Independent Variable (X)**

Regression Analysis Cont...

Dependent Variable (Y)

The variable whose value is influenced or is to be predicted is called dependent variable. A dependent variable is the outcome or result.

Independent Variable (X)

The variable which influences the values or is used for prediction is called independent variable. **The independent variable is the cause that produces the effect.**

Regression Analysis Cont...

Dependent Variable

Dependent variable is also known as –

- Regressed Variable
- Explained Variable
- Response Variable
- Outcome Variable
- Endogenous Variable

Independent Variable

Independent variable is also known as –

- Regressor
- Predictor
- Explanatory Variable
- Exogenous Variable

Advantages of Regression Analysis

- Regression analysis estimates the relationship that exists, on the average between the dependent variable and the independent variables.
- It shows the nature of relationship between two or more variables.
- It determines the effect of each of the explanatory variables on the dependent variable.
- **Regression predicts the value of dependent variable for a given value of the independent variable.**
- **Regression analysis is used for forecasting.**

Types of Regression

- Simple Regression: It involves only two variables.
- Multiple Regression: It involves more than two variables.
- Linear regression: If the regression curve is a **straight line** then there is a linear regression between the variables.
- Non-linear regression: If the regression curve is not a straight line then there is a **non-linear or curvilinear** regression between the variables

Simple Linear Regression

Simple Linear Regression

- Simple linear regression analysis is a statistical tool that gives us the ability to **estimate the mathematical relationship between a dependent variable and an independent variable.**
- Use the data to position a line that best represents the relationship between the two variables.
- If we are interested to study the **dependency of a variable on a single variable**, we use the method of simple regression.

Some Applications of Simple Linear Regression

- A scientist wants to know at which level of sound pollution will effect human health.
- A company wants to investigate the relationship between advertising spending and revenue.
- Medical researchers may want to understand the relationship between drug dosage and blood pressure of patients.
- Agricultural scientists may want to measure the effect of fertilizer and water on crop yields.
- A sale manager wants to predict the total sale in next year based on the number of staffs.

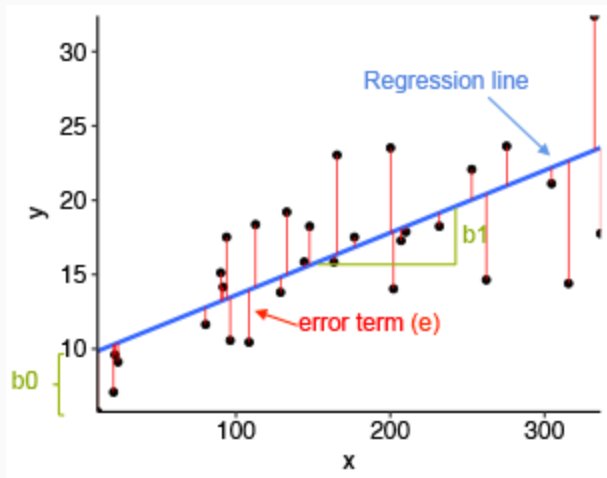
Simple Linear Regression

- Simple linear regression analysis involves **one dependent variable (y)** and **one independent variable (x)** in which the relationship between the variables is approximated by a straight line.
- The general form of a simple linear regression model is

$$y = \beta_0 + \beta_1 x + \epsilon$$

- Here, β_0 = **intercept parameter**, β_1 = **slope parameter** and ϵ (epsilon) is a random variable referred to as the error term.
- The error term accounts for the variability in y that can not be explained by the linear relationship between y and x.

Simple Linear Regression



Simple Linear Regression Equation

- Each distribution of y values has its own mean or expected value.
- Regression equations describes how the expected value of y (is denoted by $E(y)$) is related to the independent variable x .
- This equation is called the regression equation.

$$E(y) = \beta_0 + \beta_1 x$$

where,

- β_0 is the intercept of the regression line which measures the average value of y when $x=0$.
- β_1 is the slope which measures the average rate of change in dependent variable per unit change in independent variable.
- And $E(y)$ is the mean or expected value of y for a given value of x .

Estimated Simple Linear Regression Equation

- If the population parameters β_0 and β_1 were known the mean value of y for a given value of x can be determined.
- **In practice, the parameter values are not known and we need to estimate them.**
- Sample statistics (denoted by b_0 and b_1) are computed as estimates of the population parameters β_0 and β_1 .

$$\hat{\beta}_0 = b_0 \quad \text{and} \quad \hat{\beta}_1 = b_1$$

- The estimated regression equation for simple linear regression follows

$$\hat{y} = b_0 + b_1x$$

where, b_0 is the y intercept, b_1 is the slope and $\hat{y}$ is the point estimator of $E(y)$, the mean value of y for a given value of x .

Assumptions of Simple Linear Regression

- The relationship between the dependent and independent variables is linear.
- The error term ϵ is a random variable with a mean or expected value of zero; that is, $E(\epsilon) = 0$.
- The variance of ϵ , denoted by σ^2 is the same for all values of x .
- The values of ϵ are independent.
- The error term ϵ is a normally distributed random variable.

Estimating the parameters

- The ordinary least squares (OLS) method is the most widely used method to estimate the regression equation.
- The least squares method provides an estimated regression equation that minimizes the sum of squared deviations between the observed values of the dependent variable y and the estimated values of the dependent variable $\hat{y}$.
- The sum of squared residuals is given by $\sum (y_i - \hat{y}_i)^2$
- This least squares criterion is used to choose the equation that provides the best fit.

Residual/error = difference between the actual y and the predicted y

That means, $e_i = y_i - \hat{y}_i$

Estimating the parameters

Using ordinary least square (OLS) method we can estimate the parameters as

$$b_1 = \frac{\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}}{\sum_{i=1}^n x_i^2 - n \bar{x}^2} \quad \text{and} \quad b_0 = \bar{y} - b_1 \bar{x}$$

Interpretation of b_1

b_1 is the estimated change in the average value of y as a result of a one-unit change in x .

Interpretation of b_0

b_0 is the value of the dependent variable (y) when the value of independent variable (x) is zero.

Merits and Demerits

Merits

- Simple linear regression works well irrespective of data size.
- Simple linear regression gives information about the relevance of features.

Demerits

- If the assumptions of simple linear regression are violated then, it provides biased or inefficient estimates.
- Results are strongly affected by outliers in simple linear regression.
- It assumes that all the variables are normally distributed.

Example 1

An auto manufacturing company wanted to investigate how the price of one of its car models depreciates with age. The research department at the company took a sample of ten cars of this model and collected the following information on the ages (in years) and prices (in hundreds of dollars) of these cars.

Age	8	3	6	4	9	2	3	5	6	3
Price	45	210	100	190	33	267	190	134	109	235

Questions

1. Estimate the regression model for the above data.
2. What is your interpretation for the model?
3. What is the price of a 7 years old car? / Predict the price of a 7 years old car.
4. Find the residual for a 4 years old car.
5. Find the Coefficient of Determination and interpret?
6. Draw a scatter plot for the above data and fit the regression line in the same plot.

Solution of Example 1

Age(x)	Price(y)	xy	x^2	y^2
8	45	360	64	2025
3	210	630	9	44100
6	100	600	36	10000
4	190	760	16	36100
9	33	297	81	1089
2	267	534	4	71289
3	190	570	9	36100
5	134	670	25	17956
6	109	654	36	11881
3	235	705	9	55225
$\sum x = 49$	$\sum y = 1513$	$\sum xy = 5780$	$\sum x^2 = 289$	$\sum y^2 = 285765$

Solution of Example 1 Cont...

Here, $\bar{x} = 4.9$ and $\bar{y} = 151.3$

$$b_1 = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} = -33.41$$

$$b_0 = \bar{y} - b_1\bar{x} = 315$$

1. The estimated regression model:

$$\hat{y} = 315 - 33.41x$$

2. The interpretation for the model:

$b_0 = 315$ means the average price will be 31500\$ for a newly launched car of that model.

$b_1 = -33.41$ means that the average price will be decreased by 3341\$ when car age will be increased by 1 year.

Solution of Example 1 Cont...

3. The price of a 7 years old car:

$$\hat{y} = 315 - 33.41 \times 7 = 81.13$$

The price of a 7 years old car is 8113\$.

4. The residual of a 4 years old car:

$$\hat{y} = 315 - 33.41 \times 4 = 181.36$$

$$\begin{aligned} \text{Residual} &= y - \hat{y} \\ &= 190 - 181.36 \\ &= 8.64 \end{aligned}$$

Solution of Example 1 Cont...

5. Coefficient of determination:

$$\begin{aligned} r &= \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sqrt{(\sum_{i=1}^n x_i^2 - n\bar{x}^2)(\sum_{i=1}^n y_i^2 - n\bar{y}^2)}} \\ &= \frac{5780 - 10 \times 4.9 \times 151.3}{\sqrt{(289 - 10 \times 4.9^2)(285765 - 10 \times 151.3^2)}} \\ &= -0.98 \\ r^2 &= 0.9604 \end{aligned}$$

Interpretation: 96.04% variation in price of a car can be explained by the age of the car.

6. Draw a scatter plot for the above data and fit the regression line in the same plot.

See the class note

Example 2

The following data is extracted from an article “Reactions on Painted Steel under the Influence of Sodium Chloride and Combinations Thereof”. The independent variable (x) is SO_2 deposition rate ($mg/m^2/d$) and the dependent variable (y) is steel weight loss (g/m^2).

x	y
14	280
18	350
40	470
43	500
45	560
112	1200

Questions

1. Develop the estimated regression equation on steel weight loss given the SO_2 deposition rate .
2. What is your interpretation for the model?
3. Predict the steel weight loss with SO_2 deposition rate 50 $mg/m^2/d$.
4. Find the residuals for the fifth observation.
5. Comment on the coefficient of determination of the model.
6. Draw a scatter plot for the above data and fit the regression line in the same plot.

Solution of Example 2

```
lm(formula = y ~ x)
```

Residuals:

1	2	3	4	5	6
11.762	44.516	-40.338	-38.273	3.104	19.229

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	137.8756	26.3776	5.227	0.0064 **
x	9.3116	0.4745	19.622	3.98e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 37.39 on 4 degrees of freedom

Multiple R-squared: 0.9897, Adjusted R-squared: 0.9871

F-statistic: 385 on 1 and 4 DF, p-value: 3.978e-05

1. The estimated regression model:

$$\hat{y} = 137.8756 + 9.3116x$$

2. The interpretation for the model:

$b_0 = 137.8756$ means average steel weight loss will be 137.8756 g/m^2 , when the SO_2 deposition rate is zero.

$b_1 = 9.3116$ means the average steel weight loss will increase by 9.3116 g/m^2 when the SO_2 deposition rate will increase by one unit.

Solution of Example 2 Cont...

3. The steel weight loss with SO_2 deposition rate 50 $mg/m^2/d$ is

$$\hat{y} = 137.8756 + 9.3116 \times 50 = 603.4556$$

So, the predicted steel weight loss is approximately 603.4556 g/m^2 .

4. The residual for the fifth observation:

$$\hat{y} = 137.8756 + 9.3116 \times 45 = 556.8976$$

$$\begin{aligned} \text{Residual} &= y - \hat{y} \\ &= 560 - 556.8976 = 3.1024 \end{aligned}$$

5. Coefficient of determination: $r^2 = 0.9871$

Interpretation: 98.71% variation in steel weight loss can be explained by SO_2 deposition rate.

Test Yourself

Father's Height (inches)	Son's Height (inches)
58	60
60	62
62	63
64	64
65	64
65	64
66	65
67	66
70	67
73	69
75	70

Questions

1. Establish the relationship between Fathers Height and Sons Height ?
2. Fit a regression line of Sons Height on Fathers Height.
3. What will be Sons Height if the Fathers Height is 74 inches?
4. Find coefficient of determination and find how much variation of Sons Height is explained by the Fathers Height .
5. What is your interpretation for the model?

Practice Problem

Textbook: Probability and Statistics for Engineering and the Sciences (Jay L. Devore)

CHAPTER 12: Simple Linear Regression and Correlation

Page 488: 17, 19

References and Acknowledgement

- Reference Books:

1. Introductory Statistics by Neil A. Weiss, Ph.D.
2. INTRODUCTORY STATISTICS by PREM S. MANN

- Acknowledgement:

1. Dr. Mohammad Rafiqul Islam, Professor, MNS, Brac University.
2. Md. Ariful Islam Sanim, Lecturer, MNS, Brac University.

Thank You